



08: —

Phase 2 — | 45

Part A:

1: —

$$3x + 5 > 2x - 1.$$

① $x \rightarrow 3x - 2x + 5 > -1.$

② $5 \rightarrow x > -6.$

$$-2x < 6.$$

① $-2 \rightarrow$

② $\rightarrow x > -3.$

$$-3 < 2x + 1 \leq 5.$$

① $1 \rightarrow -4 < 2x \leq 4.$

② $2 \rightarrow -2 < x \leq 2.$

$-2, 2, .$

2: —, ,

$$x^2 - 5x + 6 > 0.$$

① $(x - 2)(x - 3) > 0.$

② $0: x = 2, x = 3.$

③ $.$

- $$\rightarrow 1 < x \leq 3.$$

$$x^3 - 3x^2 - 4x + 12 \geq 0.$$

① $x = 2 \diamond ? \diamond ? \diamond ? \diamond ? : 8 - 12 - 8 + 12 = 0. \diamond 0.$

$$\textcircled{2} \text{ ? ? ? ? ? ? } (x - 2) \text{ ? ? ? ? ? .}$$

$$\diamondsuit x^2 - x - 6 = (x - 3)(x + 2).$$

$$\rightarrow (x - 2)(x - 3)(x + 2) \geq 0.$$

③ $0 \diamond ? \diamond ? \diamond ? : x = -2, 2, 3. \diamond ? \diamond ? \diamond ?$.

- $$\rightarrow -2 \leq x \leq 2 \quad \diamond \diamond \quad x \geq 3.$$

$$(x - 1)^2(x + 2) < 0.$$

$$\textcircled{1} \, 0 \, \diamond \, \diamond \, \diamond \, \diamond : x = 1 \, (\diamond \, \diamond - \diamond \, \diamond), x = -2 \, (\diamond \, \diamond - \diamond \, \diamond).$$

(2) ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? . ? ? ? ? ?

③ $\diamondsuit \diamondsuit :$

- $x < -2$: $(+)^2(-) = -$. ✓
- $-2 < x < 1$: $(+)^2(+) = +$. ✗

- $x > 1: (+)^2(+)=+.$ ✕

$$\rightarrow x < -2.$$

9: —

$$|x - 3| < 5.$$

① "0".

$$-5 < x - 3 < 5.$$

② $3 \rightarrow -2 < x < 8.$

$$|2x + 1| \geq 3.$$

① 0:

$$2x + 1 \leq -3 \quad 2x + 1 \geq 3.$$

② $2x \leq -4 \rightarrow x \leq -2. \quad 2x \geq 2 \rightarrow x \geq 1.$

$$\rightarrow x \leq -2 \quad x \geq 1.$$

10: —

$$|x - 1| + |x + 2| \leq 5.$$

① 0: $x = 1, x = -2.$

② .

1: $x < -2$

$$\rightarrow .$$

$$-(x - 1) - (x + 2) = -2x - 1 \leq 5.$$

$$-2x \leq 6 \rightarrow x \geq -3.$$

$$\rightarrow -3 \leq x < -2.$$

2: $-2 \leq x < 1$

$$(x - 1), (x + 2).$$

$$-(x-1) + (x+2) = 3 \leq 5.$$

$$\rightarrow -2 \leq x < 1.$$

$$\textbf{3: } x \geq 1$$

$$\rightarrow$$

$$(x-1) + (x+2) = 2x+1 \leq 5.$$

$$2x \leq 4 \rightarrow x \leq 2.$$

$$\rightarrow 1 \leq x \leq 2.$$

$$\textcircled{3} \rightarrow -3 \leq x \leq 2.$$

Part C:

$$\textbf{11: } -$$

$$2^{x+1} > 8.$$

$$\textcircled{1} 8 = 2^3.$$

$$\textcircled{2} 2^{x+1} > 2^3. \rightarrow 2 > 1 \rightarrow x+1 > 3 \rightarrow x > 2.$$

$$\left(\frac{1}{2}\right)^{x+1} \geq \frac{1}{4}.$$

$$\textcircled{1} \frac{1}{4} = \left(\frac{1}{2}\right)^2.$$

$$\textcircled{2} \frac{1}{2} < 1 \rightarrow x+1 \leq 2 \rightarrow x \leq 1.$$

$$\textbf{12: } > 0$$

$$\log_2(x-1) < 3.$$

$$\textcircled{1} 3 = \log_2 8.$$

$$\textcircled{2} \log_2(x-1) < \log_2 8. \rightarrow 2 > 1 \rightarrow x-1 < 8 \rightarrow x < 9.$$

$$\textcircled{3} \diamond \diamond \diamond \diamond: x - 1 > 0 \rightarrow x > 1.$$

$$\rightarrow 1 < x < 9.$$

$$\log_{\frac{1}{2}}(x + 2) \geq 1.$$

$$\textcircled{1} 1 = \log_{\frac{1}{2}} \frac{1}{2}.$$

$$\textcircled{2} \diamond \frac{1}{2} < 1 \rightarrow \diamond \diamond \diamond \diamond \diamond \diamond: x + 2 \leq \frac{1}{2} \rightarrow x \leq -\frac{3}{2}.$$

$$\textcircled{3} \diamond \diamond \diamond \diamond: x + 2 > 0 \rightarrow x > -2.$$

$$\rightarrow -2 < x \leq -\frac{3}{2}.$$

Part D: $\diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond$

$$\diamond \diamond \mathbf{13:} [x] \diamond \diamond \diamond \diamond \diamond \diamond \diamond$$

$$[x] \diamond x \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond.$$

$$\diamond \diamond \diamond \diamond \diamond \diamond:$$

$$[3.7] = 3, [3.0] = 3, [0.5] = 0.$$

$$[-1.2] = -2, [-1.0] = -1, [5] = 5.$$

$$\diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond. [3.7] = 3 \diamond \diamond 3 \leq 3.7 < 4.$$

$$[0.5] = 0 \diamond \diamond 0 \leq 0.5 < 1. [-1.2] = -2 \diamond \diamond -2 \leq -1.2 < -1.$$

$$\diamond [x] = 3 \diamond \diamond \rightarrow 3 \leq x < 4.$$

$$[2x] = 5 \rightarrow 5 \leq 2x < 6 \rightarrow 2.5 \leq x < 3.$$

$$\diamond \diamond \mathbf{14:} [x] \diamond \diamond \diamond \diamond \diamond \diamond \diamond - t \diamond \diamond \diamond \diamond$$

$$[x]^2 - [x] - 6 < 0.$$

$$\textcircled{1} [x] \diamond t \diamond \diamond \diamond \diamond : t^2 - t - 6 < 0.$$

$$\textcircled{2} \diamond \diamond \diamond : (t - 3)(t + 2) < 0 \rightarrow -2 < t < 3.$$

$$\textcircled{3} t = [x] \diamond \diamond \diamond \diamond . \rightarrow t = -1, 0, 1, 2.$$

$$\textcircled{4} \diamond t \diamond \diamond x \diamond \diamond \diamond \diamond \diamond \diamond \diamond :$$

$$[x] = -1 \rightarrow -1 \leq x < 0.$$

$$[x] = 0 \rightarrow 0 \leq x < 1.$$

$$[x] = 1 \rightarrow 1 \leq x < 2.$$

$$[x] = 2 \rightarrow 2 \leq x < 3.$$

$$\rightarrow -1 \leq x < 3.$$

$$\diamond \diamond \textbf{15: } [x + 1] > [x] - \diamond \diamond \diamond \diamond \diamond ?$$

$$[x + 1] > [x].$$

$$x \diamond \diamond \diamond n \diamond \diamond : [n + 1] = n + 1, [n] = n \rightarrow n + 1 > n. \diamond.$$

$$x \diamond \diamond \diamond \diamond \diamond \diamond , x = n + \delta : [x + 1] = n + 1, [x] = n \rightarrow n + 1 > n. \diamond.$$

$$\rightarrow \diamond \diamond x \diamond \diamond \diamond \diamond . \diamond \diamond : [x + 1] = [x] + 1 \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond .$$

$$\diamond \diamond \textbf{16: } \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond$$

$$[x] : x \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond .$$

$$[3.2] = 4, [-1.2] = -1.$$

$$\{x\} = x - [x] : x \diamond \diamond \diamond \diamond \diamond . \diamond \diamond 0 \leq \{x\} < 1.$$

$$\{3.7\} = 0.7, \{-1.2\} = 0.8.$$

$$\diamond \diamond :$$

$$[x] = [x] \rightarrow x \diamond \diamond \diamond \diamond \diamond .$$

$$[x] = [x] + 1 \rightarrow x \diamond \diamond \diamond \diamond \diamond \diamond \diamond .$$

?? 4: ??

?? ?:"(x-1)²(x+2)?? x=1?? ?? ?? ?? ?? ??."

?? ???:(x-1)²? x=1?? 0?? ??, ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ??.
x=1 ?? ?? ?? (x-1)²>0. ?? ?? ?? ??.

?? ??:(x-a)²?? ?? ?? ??, (x-a)²?? ?? ?? ?? ??.

?? ?? ?? ?? ?? ??

- ① ?? ?? 0?? ?? ???. ?? ?? ?? ?? ??.
- ② 0? ?? ?? ?? ?? ?? ???. ??=0? ?? ?? ?? ??.
- ③ ? ?? ?? ?? ? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ??.
- ④ ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ??.
- ⑤ ??·?? ??>1?? ?? ?? ??, 0<?<1?? ?? ??.
- ?? ?? ? ??>0 ?? ?? ?? ?? ??.
- ⑥ ?? ?? 0 ?? ?? ?? ?? ?? ?? ?? ?? ??.
- ⑦ [x]? t=[x] ?? ???. t ?? ?? ?? ?? ?? ??.

?? ? 1

$\frac{x^2-4}{x+3} \leq 0$. ?? ?? ??, ?? ≠ 0 ??, ?? ?? ?? ?? ?? ??.

→ ?? ???: ?? 5

???: ?? ??

?? ? 2

$2^x + 2^{x+1} > 48$. 2^x ?? ?? ?? ?? ?? ?? ?? ??.

→ : 11

:

3

$\log_2(x^2 - 3x) \leq 2.$ > 0 .

→ : 12

:

4:

$[x]^2 - 5[x] + 6 = 0$ x
 $[x]$.

→ : 14

:

5

$|x - 2| + |x + 3| \leq 7.0$.

→ : 10

:

6:

$\{x\} + [x]^2 \leq x \leq 0 \leq x < 3$
 $\{x\} = x - [x]$

→ 14, 16

1: 0
2: 0
(
3:
- :
- · : >1, 0<<1. >0
- n: 0
- [x]: t=[x], t

"", "", "", "", ""
.

		<1
0		critical point
		interval method
		sign chart

$\diamond \diamond \diamond \diamond \diamond \diamond \diamond$	$\diamond \diamond \diamond \diamond \diamond$	$\diamond \diamond \diamond / \diamond \diamond$
$\diamond \diamond \diamond \diamond$	$\diamond \diamond \diamond \diamond$	$D = b^2 - 4ac$
$\diamond \diamond$	$\diamond \diamond$	$\diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond$
$\diamond$	$\diamond$	base
$\diamond \diamond \diamond \diamond \diamond$	$\diamond \diamond \diamond \diamond \diamond (\diamond \diamond \diamond \diamond \diamond \diamond \diamond)$	$\lfloor x \rfloor = \lceil x \rceil$
$\diamond \diamond \diamond \diamond \diamond$	$\diamond \diamond \diamond \diamond \diamond$	$\lceil x \rceil$
$\diamond \diamond \diamond \diamond \diamond$	$\diamond \diamond \diamond \diamond$	$\{x\} = x - \lfloor x \rfloor$
$\diamond \diamond \diamond \diamond$	$\diamond \diamond \diamond \diamond$	$ x $